Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam E · Form 1: Constraint analysis

Multiple Choice Questions: Select the best answer. (1 point each)

1. In projectile video, horizontal velocity is constant; every recorded vertical coordinate is shifted upward by the same distance. Which estimate is preserved?

(A) Launch height relative to a zero floor

(B) Recorded apex coordinate

(C) g inferred from curvature of y(t)

(D) Initial vertical intercept

2. A target is 4 m horizontally away and 1.225 m below a horizontal launcher. Ignore drag. Find the required horizontal launch speed. Use g = 9.8 m/s².

(A) -8 m/s

(B) 16 m/s

(C) 4 m/s

(D) 8 m/s

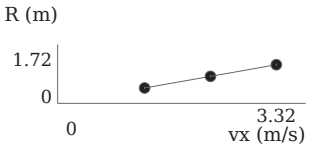
3. A ball launches horizontally from a 1.25 m table. Use g=10 m/s² and neglect drag. vx (m/s) -> R (m): (1.32, 0.66); (2.32, 1.16); (3.32, 1.66). At fixed sensor resolution, how can slope uncertainty be reduced?

(A) Use a wider horizontal baseline within the valid regime.

(B) Shorten the baseline.

(C) Append digits to one reading.

(D) Discard the sign of every reading.



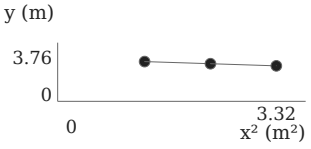
4. A ball leaves a 4 m ledge horizontally at 5 m/s. Use g=10 m/s² and consider airborne frames only. Double horizontal launch speed at the same height. At the same X=2.32, what is the new minus old vertical reading?

(A) 0.348

(B) 0.39

(C) 0.264

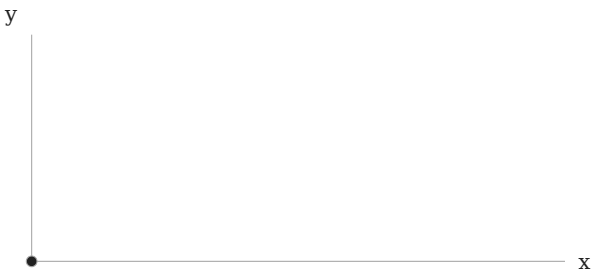
(D) 0.474



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A ball is launched from ground at 24 m/s, 40° above horizontal, and lands on the same ground. Ignore drag.

a. Draw the initial velocity vector and its horizontal and vertical components. **(1 pt)**



b. Find flight time, range, and velocity and acceleration at the top. **(3 pt)**

6. Launch at 30° and land at launch height, without drag. Use g=10 m/s². u^2 (m^2/s^2) -> R (m): (1.32, 0.1143); (2.32, 0.2009); (3.32, 0.2875).

a. Build the model for level-ground range from physics and predict both slope and intercept. **(2 pt)**

b. Use 60° while preserving speed and equal launch and landing heights. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A projectile starts at (0,h), at speed u and angle θ above horizontal, with cosθ > 0. Eliminate time to derive the trajectory y(x). **(3 pt)**